PILLAR: Climate Change

Our Approach

Addressing climate change risks to the Company and reducing our impacts on the climate are strategic priorities for Lundin Gold, which we manage through risk mitigation and management, policy awareness, and leveraging opportunities. By employing the Task Force for Climate-Related Financial Disclosures (TCFD) framework, we aim to continually understand our climate risks, opportunities, and responsibilities, evaluate their potential impacts on our business through scenario analysis, develop robust climate-related governance and strategies, and measure, monitor and disclose our progress through a TCFD-aligned Climate Report.

5-Year Strategy

***OUR 5-YEAR OBJECTIVE** is to reduce our global emissions footprint and become industry leaders in reducing the emissions intensity of gold mining in Ecuador.*

What we hope to accomplish by 2025

- Establish Lundin Gold as a leader in efforts to reduce the GHG emission intensity of gold mining in Ecuador.
- By the end of 2022, establish our operational emissions baseline and set an intensity-based target commensurate with our commitment to lead on this issue. In subsequent years, we will take concrete actions to meet our target.
- Seek out innovative offset opportunities and cross-sector partnerships that have climate-related benefits.
- Identify key risks we face from a changing climate and take appropriate action to mitigate such risks. We will consider establishing an appropriate internal carbon price that we integrate into financial evaluations.
- Work with suppliers and customers to identify opportunities to reduce GHG emissions in our value chain.

Machinaza river, near Las Peñas Camp

- Transparently communicate our climate change strategy and key actions to external stakeholders, including through the incorporation of annual TCFD reporting and alignment with SDG 13 (Climate Action).
- Ensure that the Company's Board and Executive Management team have developed the expertise to ensure that this issue is well managed.

How we plan to get there

- Finalize a clear climate vision and engage with our workforce to promote this vision.
- Conduct training for Executive Management and the Board.
- Measure and report our Scope 1 and Scope 2 emissions. Where possible, put in place a strategy to understand and report Scope 3 emissions, as well.
- Identify areas for direct emission reductions (e.g., power mix, efficiency opportunities, etc.).
- Identify potential offset opportunities in communities and areas close to FDN. Engage with other Lundin Group companies to coordinate possible joint initiatives.
- Build climate resilience within our supply chain.

- Engage with upstream and downstream stakeholders to understand their climate change commitments and how Lundin Gold could possibly align with them.
- Engage with other climate leaders to remain at the forefront of this issue (e.g., local and national governments, NGOs, international agencies, etc.).
- Engage with the investment community to better understand climate reporting and disclosure expectations.

Key Headline Target

- In 2022, we will complete the measurement of our GHG emissions for the year 2021. This will be our baseline year for target setting.
- Establish an emissions intensity target and high-level plan to achieve this target in 2022.

Our Performance

TCFD Reporting

In 2021, Lundin Gold took critical steps to establish solid foundations for our climate change governance and strategy through the advancement of our TCFD reporting. Building on the election of Dr. Gillian Davidson to Lundin Gold's Board of Directors, thereby increasing the expertise of the Board on sustainability and climate change, we restructured our Environment, Health, and Safety (EHS) Committee to form the new Health, Safety, Environment and Sustainability Committee (HSES). While the purpose of the new HSES Committee is to address all sustainability matters, the Committee mandate purposefully regularizes the reporting of climate-related risks to the Board and assigns responsibility for climate risk oversight. At the same time, a decision was taken to modify both the Board Mandate to specifically address oversight of climate change and the CEO job description to expressly place ultimate responsibility for the management of climate-related risks, opportunities, and strategy in the hands of the CEO.

With the guidance of a specialized climate consultant, we began the implementation of our strategic plan within the climate change pillar of our 5-Year Sustainability Strategy. This included conducting several facilitated climate change workshops for our Board and Executive Management throughout the year so as to increase climate change literacy and thus enable more effective identification, management, and oversight of climate-related risks and opportunities. Through the evaluation of three carefully selected climate change scenarios, we considered what the future could hold for Lundin Gold. We developed leading indicators for

each scenario to help determine the most likely scenario and consider associated risks and opportunities, which will be integrated into our strategic enterprise risk management process. This will facilitate the development of short-term and long-term mitigation plans across all relevant departments of the Company.

Our 2021 Individual Performance Measures for executive compensation incorporated the achievement of key TCFD progress goals (*see Information Circular dated 16 March 2022 for more details*).

Plans for 2022 include: further aligning our executive compensation structure with the TCFD plan, establishing a Climate Resilience Group to develop Lundin Gold's approach to decarbonization and adaptation, and formulating a life-of-mine emissions intensity profile for FDN to assist us with setting informed targets.

In May 2022, we intend to issue our inaugural standalone TCFD-aligned Climate Report, which will provide additional details about our TCFD plans and progress.

GHG Emissions and Energy Consumption

Tracking and reporting of our Scope 1 and 2 GHG emissions, along with elements of our Scope 3 emissions began in 2017 during the construction phase of FDN so we could begin to develop knowledge around our emissions and impact. With an uninterrupted year of operations, we have established 2021 as our baseline year for the measurement and comparison of our GHG emissions. This baseline will also be used as a basis for the development of climate change-related targets and strategies in 2022 and beyond.

Lundin Gold's total calculated Scope 1 and 2 GHG emissions in 2021 were approximately 49,569 tonnes of CO_2 equivalent (tCO_2e), which are related to our FDN operation, our offices in Quito, Los Encuentros and Vancouver, and also our exploration Program. In addition, this year we expanded the estimate of our Scope 3 data collection and reporting to better understand our broader impact throughout the value chain for informational purposes.

Scope 1 emissions are based on direct emissions from the use of diesel, gasoline, liquid propane gas, helicopter fuel and explosives for blasting at the site. Our Scope 2 emissions are from the use of grid electricity which comes from 85% and 98% renewable sources such as hydro-electric power generation in Ecuador and Vancouver, respectively.

For our Scope 3 emissions this year, we expanded our reporting from the air travel associated with employee commuting to and from FDN reported in previous years to include these additional, third-party emissions associated with FDN's upstream and downstream activities:

1. *Cement consumption at FDN*
2. *Land and air transportation of doré produced*
3. *Land and water-based transportation of concentrate produced*
4. *Selected land transportation related to the supply of materials to site*
5. *Selected land transportation of personnel outside FDN*
6. *Business air travel*
7. *Waste management*
8. *Purchased goods and services for operations*
9. *Processing of sold doré*

We are committed to improving our understanding and informing of our Scope 3 emissions and consider them in our climate strategy in the future by encouraging our suppliers, contractors, and customers to reduce our collective emissions and impacts.

All values are reported in tCO_2e emissions and include carbon dioxide (CO_2), methane (CH_4) and nitrous oxide (N_2O) emissions, as appropriate.

The GHG emissions intensities for FDN have been calculated per tonne of ore milled and per ounce of gold produced. With our emission baseline established using 2021 data, the next step in 2022 is to define our targets and set our five-year goals to improve on our baseline.

2021 GHG emissions (tonnes CO_2-equivalent)

	2021			
GHG Scope	**Scope 1[5,6]**	**Scope 2[7]**	**Scope 3[8,9]**	**Total**
Fruta del Norte (tCO_2e)	25,014	24,097	746,750	795,861
Exploration (tCO_2e)[1]	385	–	–	385
Ecuador Offices (tCO_2e)[2]	–	73	–	73
Vancouver Offices (tCO_2e)[4]	–	0.1	–	0.1
Total (tCO_2e)	**25,399**	**24,170**	**746,750**	**796,319**
% Of Total Annual CO_2e	3%	3%	94%	100%
Kilotonnes of Ore Milled (Kt)	1,415.63			
GHG Emissions Intensity (tCO_2e/Kt ore milled)[3]	**35.01**			
Ounces of Gold Produced (oz)	428,514			
GHG Emissions Intensity (tCO_2e/oz. Au produced)[3]	**0.12**			

[1] Exploration emissions consist of emissions related to fuel consumption of exploration equipment.
[2] Ecuador offices emissions consist of emissions from the Quito and Los Encuentros offices, the Las Peñas Camp, and our FDN nursery.
[3] GHG intensity is calculated based on FDN Scope 1 and Scope 2 emissions only.
[4] Emissions from the Vancouver offices consist of emissions generated by the consumption of electrical energy.
[5] Environmental Protection Agency (EPA) Emissions Factors for GHG Inventories, 21 April 2021 were used for Global Warming Potential (GWP) rates and emission factors for off-road construction / mining equipment and trucks, as well as jet fuel, gasoline stationary equipment, and Liquid Petroleum Gas (LPG) (butane and propane).
[6] Mining Association of Canada (MAC) Towards Sustainable Mining Energy and Greenhouse Gas Emissions Management Reference Guide, June 2014 was used for emissions factors for explosives, diesel and gasoline light-duty trucks, diesel for passenger cars, medium/heavy-duty trucks, and diesel-fired stationary equipment.
[7] Comisión Técnica de determinación de Factores de Emisión de Gases de efecto invernadero – CTFE, Factor de Emisión de CO_2 del Sistema Nacional Interconectado de Ecuador, Informe 2020 was used for grid-electricity emission factors and for Vancouver offices British Columbia's Grid Electricity GHG Emission Intensity factor for 2021.
[8] Scope 3 emissions include cement consumption at FDN, land and air transportation of doré produced, land and water-based transportation of concentrate produced, approximately 90% of the land transportation related to the supply of materials to site and air travel associated with employee commuting to and from FDN.
[9] Estimates for Scope 3 CO_2e emissions were based on emission factors of 1) EPA Emissions Factors for GHG Inventories, 21 April 2021, 2) MAC Towards Sustainable Mining Energy and Greenhouse Gas Emissions Management Reference Guide, June 2014., 3) GHG intensity from cement supplier, 4) Greenhouse gas reporting: conversion factors 2021 from UK Government, 5) WRI emission factors - Quantis GHG Tool, and 6) Annual global gold market GHG emissions from "Gold and climate change: Current and future impacts" from World Gold Council.

***Note:** Figures are rounded.*

2021 Total Energy Consumption by Type (GJ)

Energy Source	**GJ[1,2]**	**% Of Total**
Diesel	351,273	43.34%
Gasoline	2,586	0.32%
Heavy Fuel Oil	–	0.00%
Other Fuels (LPG, Jet A1)[3]	2,780	0.34%
Electricity (Renewable)	386,751	47.71%
Electricity (Non-Renewable)	67,177	8.29%
Total	**810,567**	**100.00%**
Kilotonnes of Ore Milled (Kt)	1,415.63	
Energy Intensity (GJ/Kt ore milled)	**572.58**	
Ounces of Gold Produced (oz)	428,514	
Energy Intensity (GJ/oz. Au produced)	**1.89**	

[1] Energy consumption figures include energy consumed at our FDN operation, Ecuador and Vancouver offices and exploration sites.
[2] Energy conversion factors from MAC Towards Sustainable Mining Energy and Greenhouse Gas Emissions Management Reference Guide, June 2014.
[3] LPG (70% propane, 30% butane) fuel used for heaters, Jet A1 fuel used for helicopters.

***Note:** Figures are rounded.*

Our Approach

Reliable community infrastructure is critical for the mutual success of local communities and Lundin Gold. As responsible corporate citizens, we work with the government and a range of stakeholders to promote safe, equitable access to infrastructure for local communities. We therefore have short-term and long-term plans in place to improve local roads, keep existing infrastructure well-maintained and accessible, and expand or build new infrastructure, including bridges, roads, and IT networks in conjunction with national authorities, local governments and other partners. At the same time, we continue to ensure that our own infrastructure requirements are adequately met so that we can operate efficiently, effectively, and safely.

Infrastructure projects have far-reaching benefits that can be transformational for communities. In addition to providing direct services to community members and groups, these projects create opportunities for local employment and procurement, improve access to markets for local businesses and agricultural activities, and facilitate better communication, health, education, and partnerships. Over the long-term, we endeavour to facilitate a cycle whereby the improvement of community infrastructure supports organic economic growth, which in turn provides the funds required to maintain and improve upon this community infrastructure. At the same time, we assist local governments to build capacity and access other sources of funding so that this critical infrastructure can be well-maintained and operated long into the future, even after our operations cease.

5-Year Strategy

***OUR 5-YEAR OBJECTIVE** is to invest in local community infrastructure to improve access to regional services and markets, build long-term capacity for the local management of infrastructure, facilitate the on-going management of FDN's critical road network, and to promote a culture of safe operations.*

Road maintenance